Homework 9

November 4, 2026

Problem 1

Let $\mathscr{A} \in \mathcal{L}(\mathcal{C}_n[\lambda])$ be defined as

$$\mathscr{A}[f(\lambda)] = f(\lambda + a), \qquad \forall\, f(\lambda) \in \mathcal{C}_n[\lambda],$$

for a nonzero constant $a \in \mathcal{C}$. We have found out the matrix representation A of $\mathscr{A}$ with respect to the basis $1, \lambda, \lambda^2, \ldots, \lambda^n$ in Problem 3 of Homework 5. Based on that result, please find out the minimal polynomial for A.

Problem 2

Let $\mathcal{V}$ be a unitary space over the field $\mathcal{F}$, and let the length of $\alpha \in \mathcal{V}$ be defined as

$$\|\alpha\| \triangleq \sqrt{\langle \alpha, \alpha \rangle}.$$

Prove the following two properties of the length function (Theorem 4.3, properties 3 and 4):

(1) $\|\alpha + \beta\| \leqslant \|\alpha\| + \|\beta\|$ for all $\alpha, \beta \in \mathcal{V}$;

(2) $|\langle \alpha, \beta \rangle| \leqslant \|\alpha\| \cdot \|\beta\|$ for all $\alpha, \beta \in \mathcal{V}$.

Problem 3

Let $\mathcal{V} = \mathcal{R}_n[\lambda]$. Given that

$$\langle p(\lambda), q(\lambda) \rangle \triangleq \int_{-1}^{1} p(\lambda) q(\lambda)\, d\lambda \qquad p(\lambda), q(\lambda) \in \mathcal{R}_n[\lambda].$$

(1) Please prove that the operation $\langle \cdot, \cdot \rangle$ defined above is inner product on $\mathcal{R}_n[\lambda]$;

(2) Find out the Gramian for the basis $1, \lambda, \lambda^2, \ldots, \lambda^{n-1}, \lambda^n$ of $\mathcal{R}_n[\lambda]$ with respect to the inner product defined above;

(3) Apply the Gram-Schmidt orthonormalization process to the basis $1, \lambda, \lambda^2, \ldots, \lambda^{n-1}, \lambda^n$ to obtain an orthonormal basis $\alpha_k L_k(\lambda),\ k = 0, 1, 2, \ldots, n$ for $\mathcal{R}_n[\lambda]$ with respect to the inner product defined above.

Problem 4

Prove the following two theorems in the textbook.

Theorem 4.7. An $n \times n$ matrix A is unitary if and only if its n column (row) vectors form an orthonormal basis for $\mathcal{C}^n$.

Theorem 4.8. Let $\mathscr{U} \in \mathcal{L}(\mathcal{V})$. The following statements are equivalent:

(1) $\mathscr{U}$ is unitary;

(2) $\|\mathscr{U}(\alpha)\| = \|\alpha\|$ for all $\alpha \in \mathcal{V}$;

(3) $\mathscr{U}$ maps an orthonormal basis of $\mathcal{V}$ to an orthonormal basis;

(4) the representation of $\mathscr{U}$ with respect to an orthonormal basis is a unitary matrix.

Problem 5

Let A be an $n \times n$ matrix. Prove that $A = A^* = A^2$ if and only if there exists an $n \times r$ matrix $U \in \mathcal{U}^{n \times r}$ such that $A = UU^*$, where $r = \operatorname{rank} A$.